

Design and Analysis of Algorithms

- **Course Code:** BCSE204L
- **Course Type:** Theory (ETH)
- **Slot:** A1+TA1 & & A2+TA2
- **Class ID:** VL2023240500901
VL2023240500902

A1+TA1

Day	Start	End
Monday	08:00	08:50
Wednesday	09:00	09:50
Friday	10:00	10:50

A2+TA2

Day	Start	End
Monday	14:00	14:50
Wednesday	15:00	15:50
Friday	16:00	16:50

Syllabus- Module 5

Module:5

Geometric Algorithms

4 hours

- Line Segments: Properties, Intersection, sweeping lines - **Convex Hull finding algorithms**: Graham's Scan, Jarvis' March Algorithm.

Introduction: **Applications and Examples of Geometric Algorithms:**

- **Sweep Line:**

- A sweep line is an imaginary line that systematically traverses the plane, typically from left to right.
- It is used to process geometric objects or events encountered along its path.
- As the sweep line progresses, it triggers events or operations based on the geometric features it encounters, such as intersections, overlaps, or changes in geometric configurations.

- **Convex Hull Algorithms:**

- These algorithms are employed to find the smallest convex polygon (Closed shapes formed by straight line segments) that encloses a given set of points.
- Convex hull algorithms are instrumental in tasks like collision detection, spatial planning, and pattern recognition.
- Techniques like **Graham's Scan** and **Jarvis' March Algorithm** are commonly used for this purpose.

- **Line Intersection Algorithms:**

- These algorithms focus on determining whether two lines in a plane intersect and, if so, finding their intersection point.
- They are applied in fields such as computer graphics, CAD, and geometric modeling to analyze spatial relationships and geometric configurations accurately.

Convex Hull Algorithms: (Introduction)

- **Convex Hull**

- The smallest convex set that encloses a set of points in Euclidean space, that contains all the points within it.
- It's the shape formed by enclosing the outermost points of the given set with the smallest possible convex shape. Imagine a rubber band stretched around a set of points - the enclosed area is the convex hull.
 - In 2D, it's a convex **polygon**;
 - In 3D, it's a convex **polyhedron**.

Convex Hull Algorithms: (Introduction)

- **Convex Hull Algorithms**

- Computational methods to efficiently find the convex hull of a point set.
- Take a set of points as **input** and **output** the convex hull.

Properties of Convex Hull ($CH(Q)$)

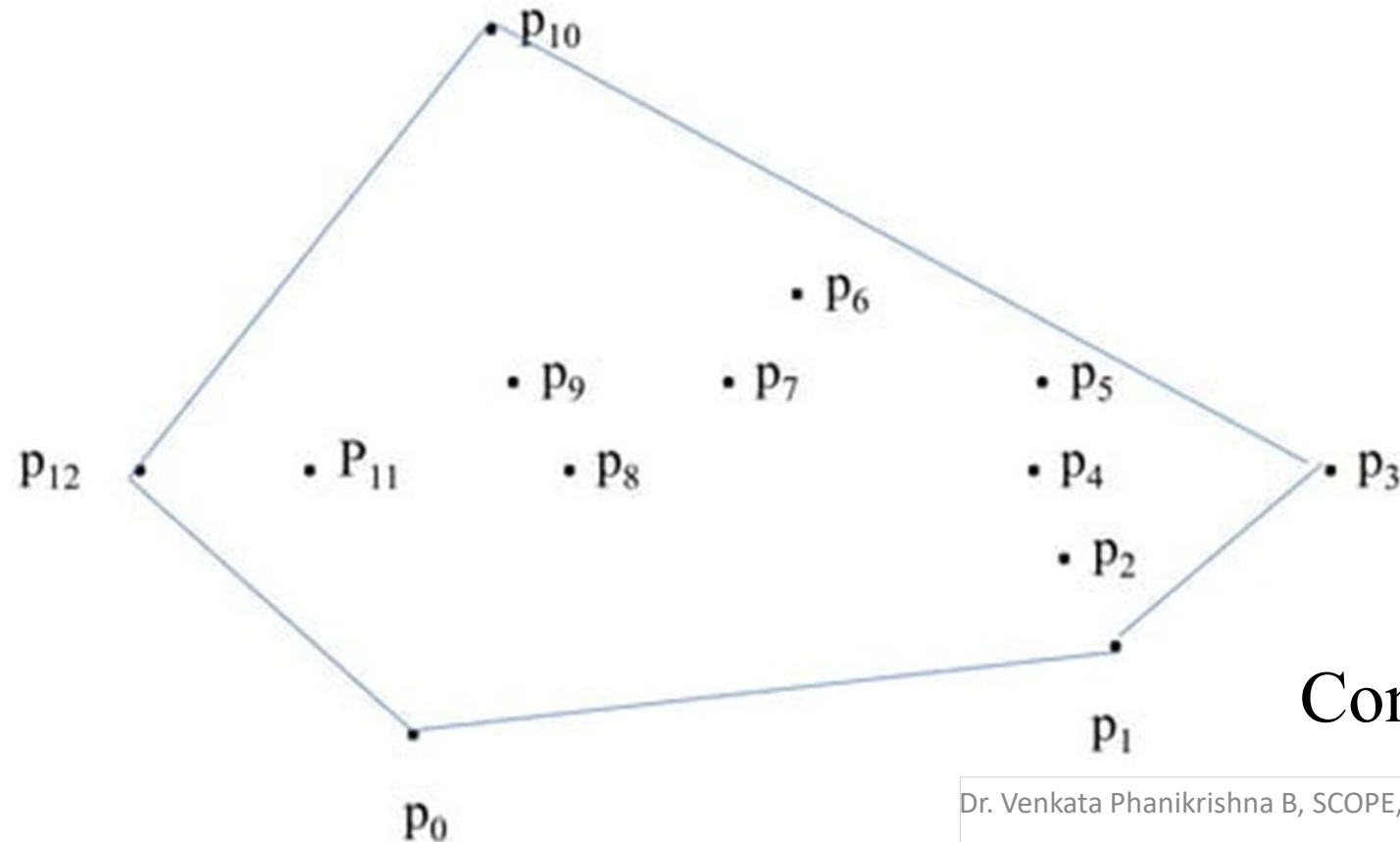
- The convex hull of a set Q of points, denoted by $CH(Q)$, is the smallest convex **polygon P** for which each point in Q is on the boundary of P or in its interior.
- The implicit assumption is that all points in set Q are unique, and Q contains **at least three points** that are **not collinear**.
- Intuitively, we can think of each point in Q as a **nail** sticking out from a board. The convex hull is then the shape formed by a tight rubber band that surrounds all the nails.

Note: "nail" is used metaphorically to represent each individual point in the set Q

Convex Hull Example:

The following figure shows a set of points and its convex hull. Points p_0 , p_1 , p_3 , p_{10} , and p_{12} represent the convex hull.

A set of points $Q = \{p_0, p_1, \dots, p_{12}\}$ with its convex hull $CH(Q)$



Convex Hull= $\{p_0, p_1, p_3, p_{10}, p_{12}\}$

Convex Hull Algorithms: (Use and Applications)

Uses of Convex Hull Algorithms:

- **Geometric processing:** Collision detection, motion planning.
- **Pattern recognition:** Shape analysis, image processing, object recognition/classification.
- **Optimization:** Defining feasible regions or constraints.

Real-world Applications:

- **Robotics/autonomous vehicles:** Path planning, obstacle avoidance, localization.
- **Computer graphics/gaming:** Rendering, collision detection, physics simulations.
- **Geographic information systems (GIS):** Spatial analysis, boundary delineation, feature extraction.
- **Image processing/computer vision:** Edge detection, object recognition, shape analysis.
- **Data visualization:** Analyze datasets in various fields (finance, biology, social sciences).

Overall, convex hulls and their corresponding algorithms are fundamental concepts in computational geometry with diverse applications across various fields.

Convex Hull Algorithms:

There are various algorithms designed to solve this problem, each with its own approach and computational complexities.

- **Incremental Method** : In this method points are first sorted from left to right, yielding a sequence $\{p_1, p_2, \dots, p_n\}$. At the i th stage we update the convex hull of the $i-1$ leftmost points, $CH(\{p_1, p_2, \dots, p_{i-1}\})$, according to the i th point from the left, thus forming $CH(\{p_1, p_2, \dots, p_i\})$. The running time of this method is **$O(n \log n)$** .
- **Divide and Conquer Method** : In this method, the set of n points are divided into two subsets in $O(n)$ time, one containing $n/2$ leftmost points and one containing $n/2$ rightmost points, recursively compute the convex hulls of the subsets. Then by means of clever method, combine the hulls in $O(n)$ time. The running time of this method is **$O(n \log n)$** .
- **Prune and Search Method** : In this method, we find the upper portion (or "upper chain") of the convex hull by repeatedly throwing out a constant fraction of the remaining points until only the upper chain of convex hull remains. Similarly we obtain lower chain. This method is asymptotically the fastest, if the convex hull contains **h vertices**, it runs in only **$O(n \log h)$** time.

Convex Hull Algorithms:

There are various algorithms designed to solve this problem, each with its own approach and computational complexities.

Polar angle : The polar angle of a point p_1 w.r.t. an origin point p_0 is the angle of the vector $p_1 - p_0$ in the usual polar coordinate system.

For example,

- The polar angle of (3, 5) w.r.t. (2, 4) is the angle of vector (1, 1) i.e. $(x_1 - x_0, y_1 - y_0)$. The polar angle $\theta = \tan(1/1) = \pi/4$ i.e. 45 degrees.
- The polar angle of (3, 3) w.r.t. (2, 4) is the angle of vector (1, -1) i.e. $(x_1 - x_0, y_1 - y_0)$. The polar angle $\theta = \tan(-1/1) = 7\pi/4$ i.e. 315 degrees.

Convex Hull Algorithms:

Rotational Sweep Method

The Rotational Sweep Method is a technique used in convex hull algorithms to efficiently compute the convex hull of a set of points. It involves sweeping a line or ray across the plane, rotating it around a fixed point, and updating the convex hull as the sweep progresses. Two popular algorithms that utilize this method are **Graham's scan** and **Jarvis's march**.

- **Graham's Scan**: This algorithm sorts the points based on their **polar angles** with respect to a reference point. It then iterates over the sorted points, maintaining a convex hull by removing concave vertices using a stack. Graham's scan has a computational time of **$O(n \log n)$** , where n is the number of input points.
- **Jarvis's March** (Gift Wrapping): Also known as the Gift Wrapping algorithm, Jarvis's march iterates over the points to find the next vertex of the convex hull. At each step, it selects the point with the smallest polar angle relative to the current vertex. Jarvis's march has a computational time of **$O(n h)$** , where h is the number of vertices on the convex hull.

Both algorithms utilize the Rotational Sweep Method to efficiently compute the convex hull while maintaining time complexity that depends on the number of points and vertices.

Convex Hull Algorithms: **Graham's Scan**

Graham's Scan Algorithm:

- The algorithm solves the convex hull problem by maintaining a **stack S** of candidate points.
- It pushes each point of the input set Q onto the stack one time, eventually popping from the stack each point that is not a vertex of $CH(Q)$.
- When the algorithm terminates, stack S contains exactly the vertices of $CH(Q)$, in counterclockwise order of their appearance on the boundary.

Convex Hull Algorithms: **Graham's Scan**

Algorithm Graham-Scan

1. Let p_0 be the point in Q with the minimum y-coordinate, or the leftmost such point in case of a tie.
2. Let $(p_1, p_2, \dots, p_m)$ be the remaining points in Q , sorted by the angles they make with the positive x-axis in counterclockwise order around p_0 (if multiple points have the same angle, keep only the one farthest from p_0).
3. Let S be an empty stack.
4. Push p_0 onto S .
5. Push p_1 onto S .
6. Push p_2 onto S .
7. For $i=3$ to m do:
While the angle formed by the points NEXT-TO-TOP(S), TOP(S), and p_i makes a non-left turn:
 POP(S).
Push p_i onto S .
8. Return S .

Note: The polar angle of a point p_1 with respect to an origin point p_0 is the angle formed by the vector from p_0 to p_1 with the positive x-axis in the usual polar coordinate system.

Convex Hull Algorithms: **Graham's Scan**

Algorithm Graham-Scan

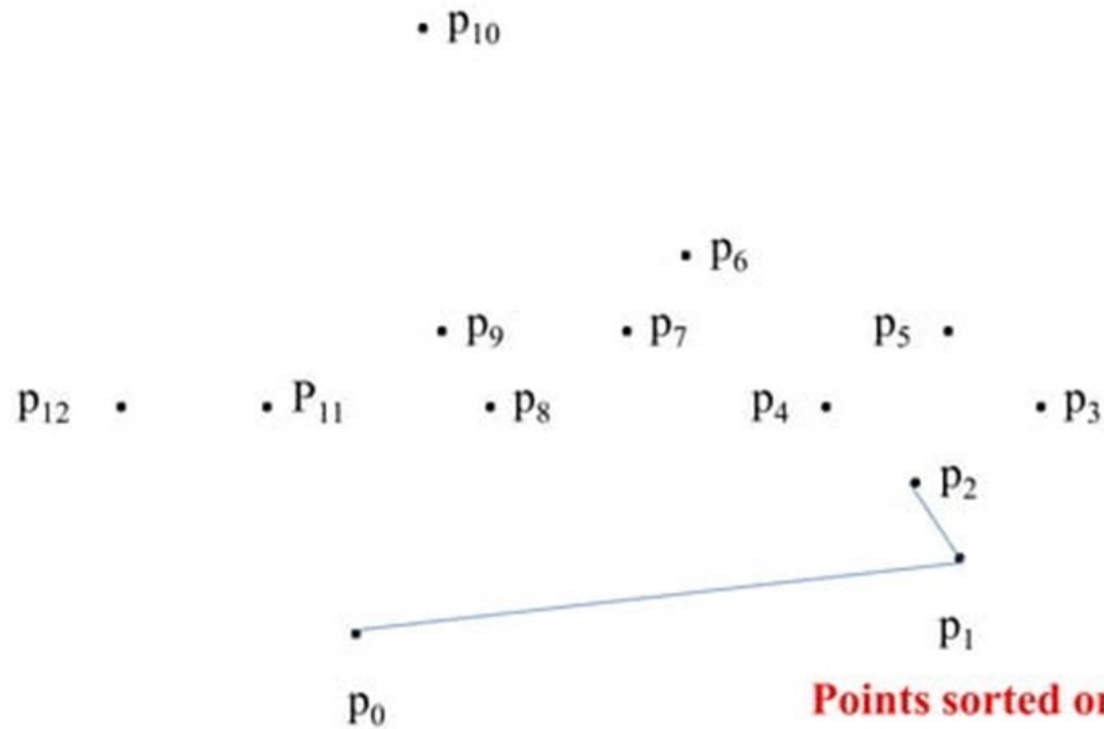
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Complexity:

- Time Complexity: $O(n \log n)$ - dominated by sorting in step 2.
- Space Complexity: $O(n)$ - due to the stack used.

Convex Hull Algorithms: **Graham's Scan** Example

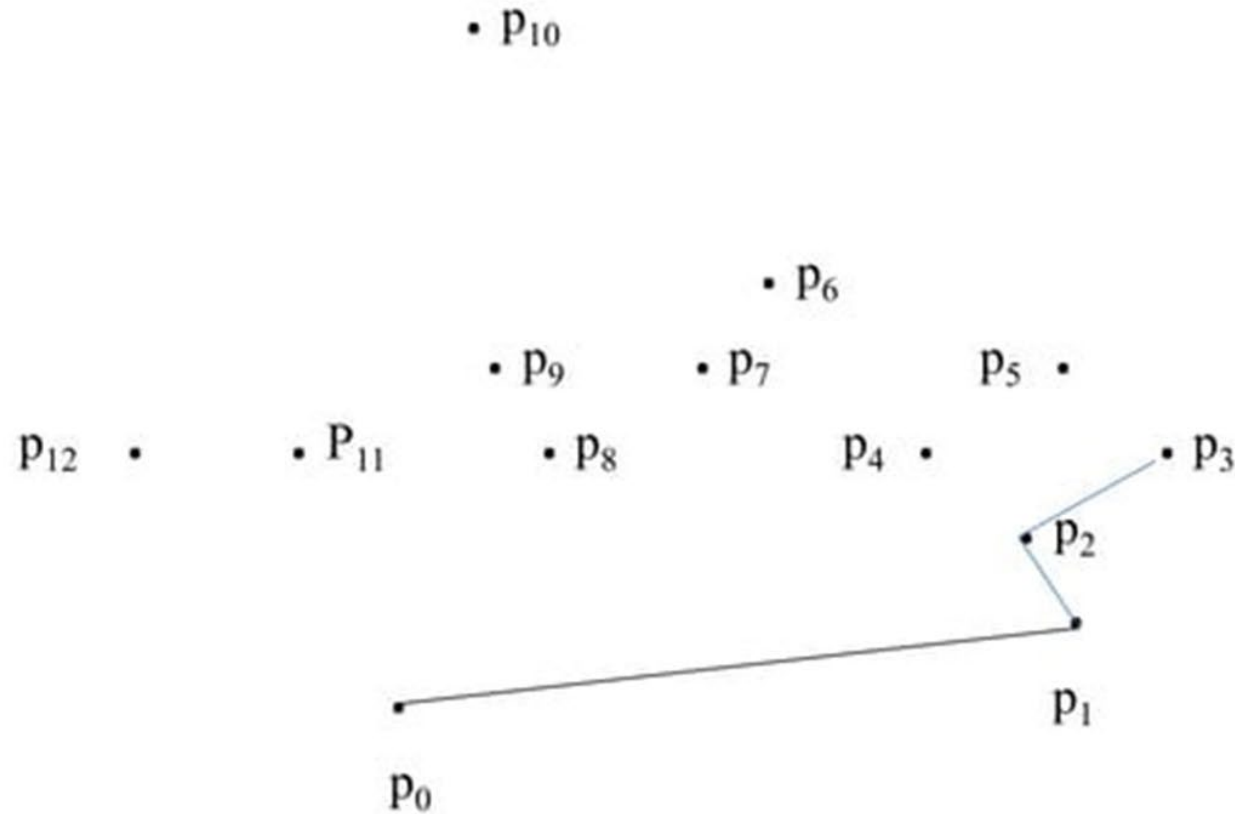
Convex Hull Algorithms : Graham's Scan Algorithm :



A set of points $Q = \{p_0, p_1, \dots, p_{12}\}$ with its convex hull $CH(Q)$

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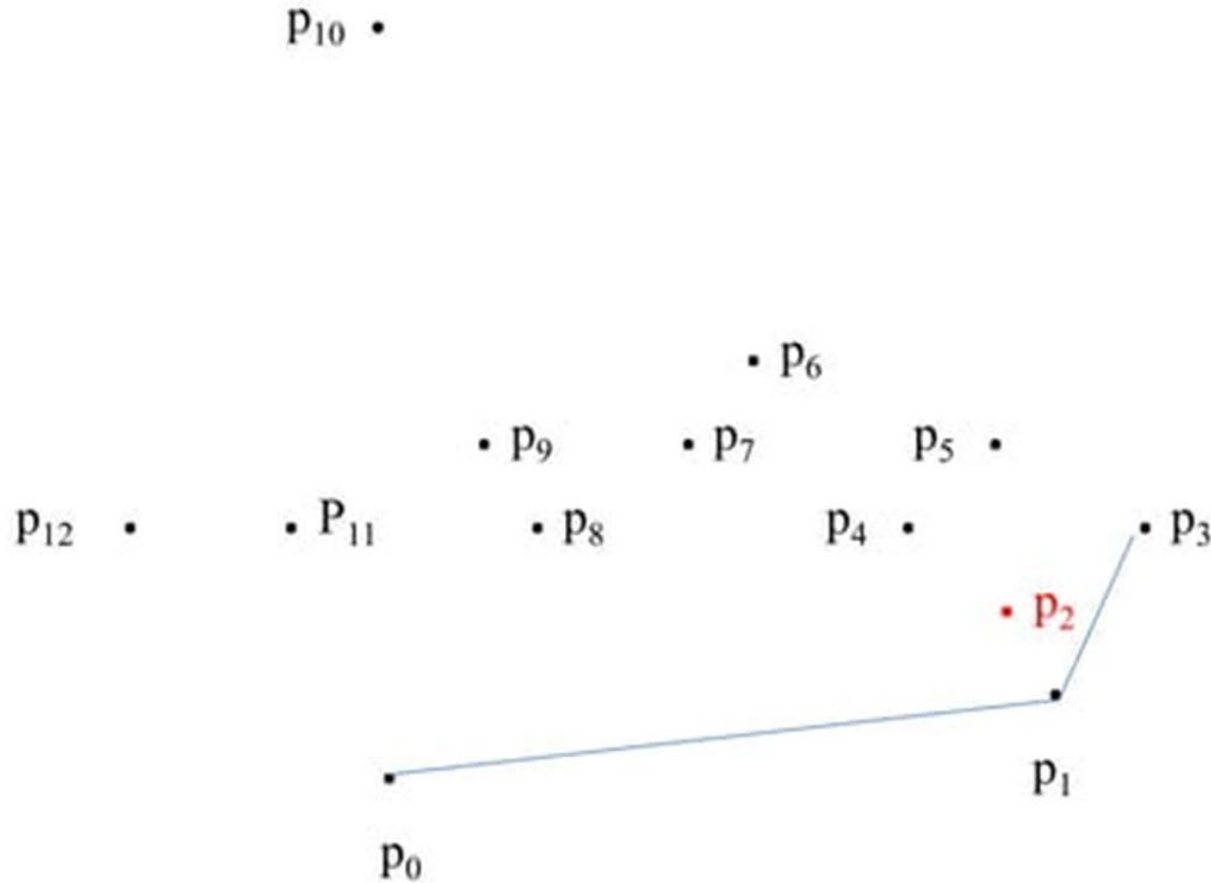
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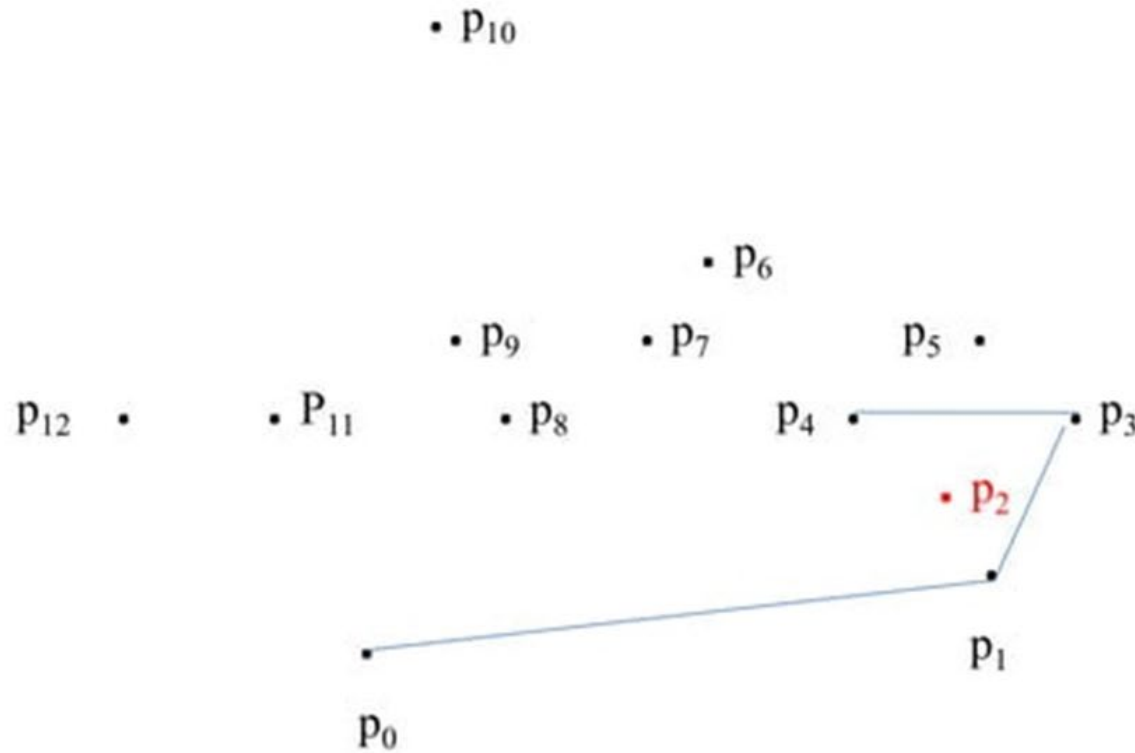
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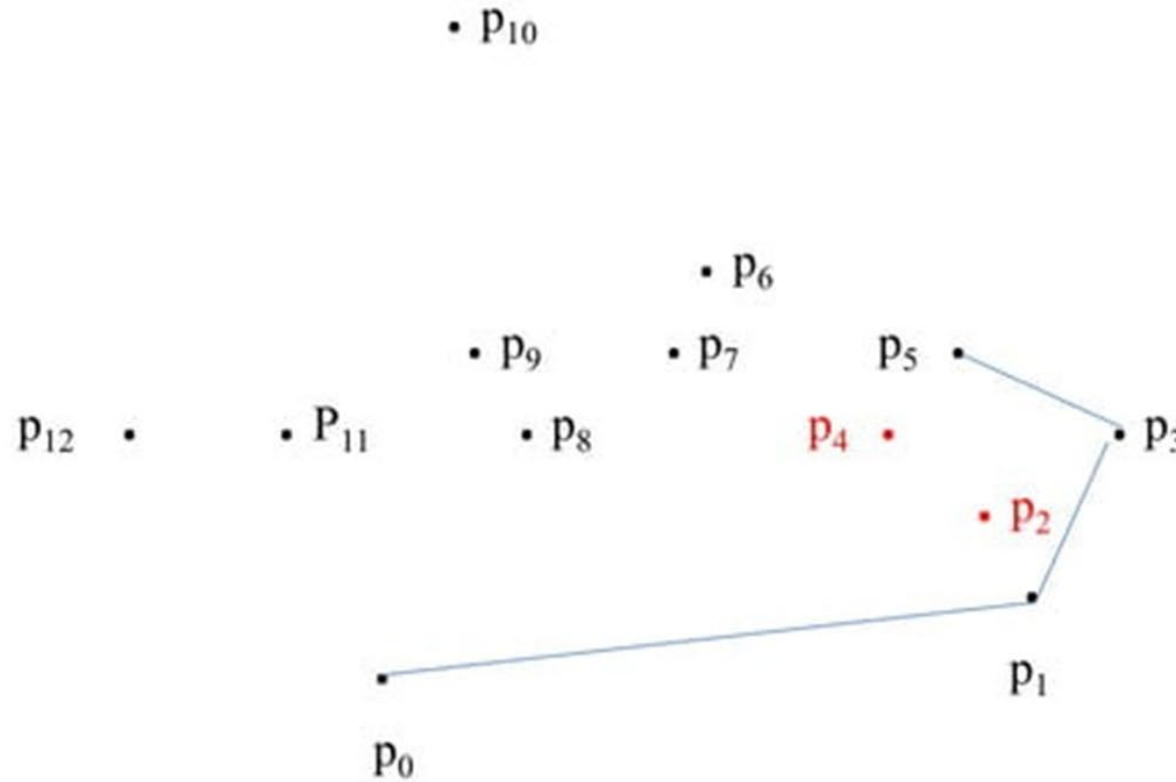
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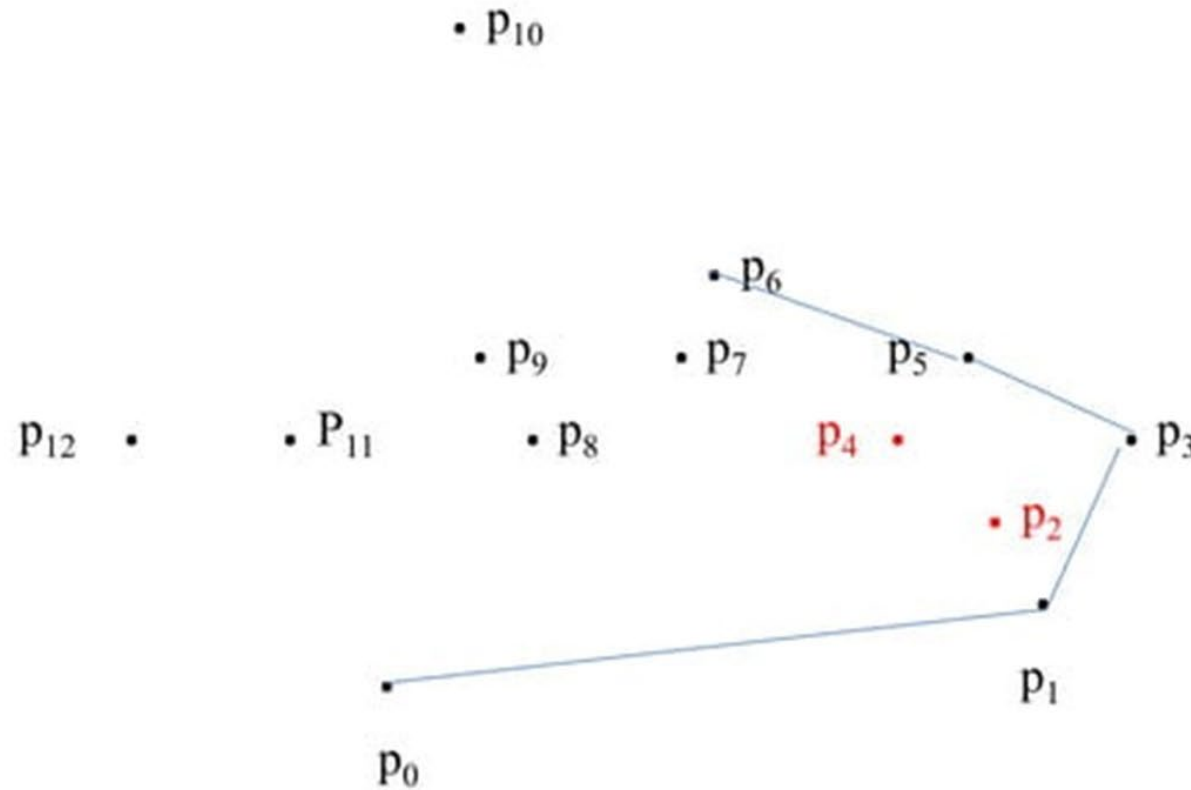
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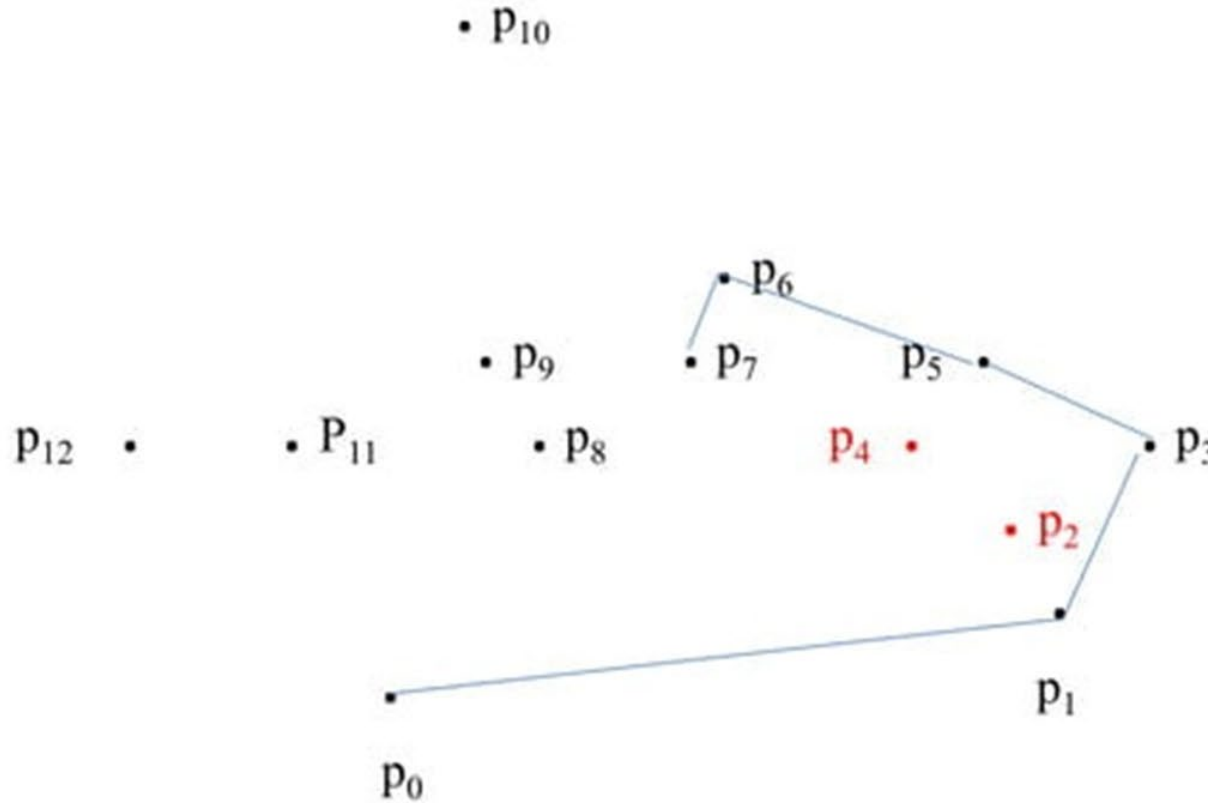
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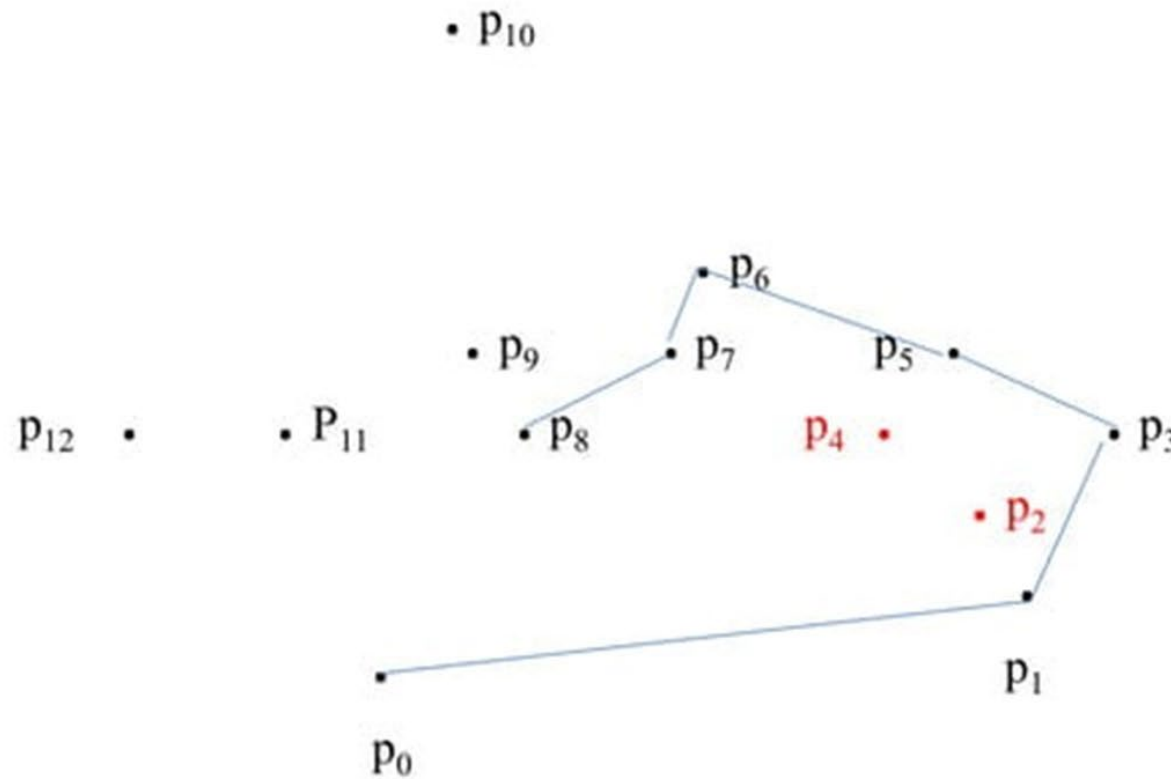
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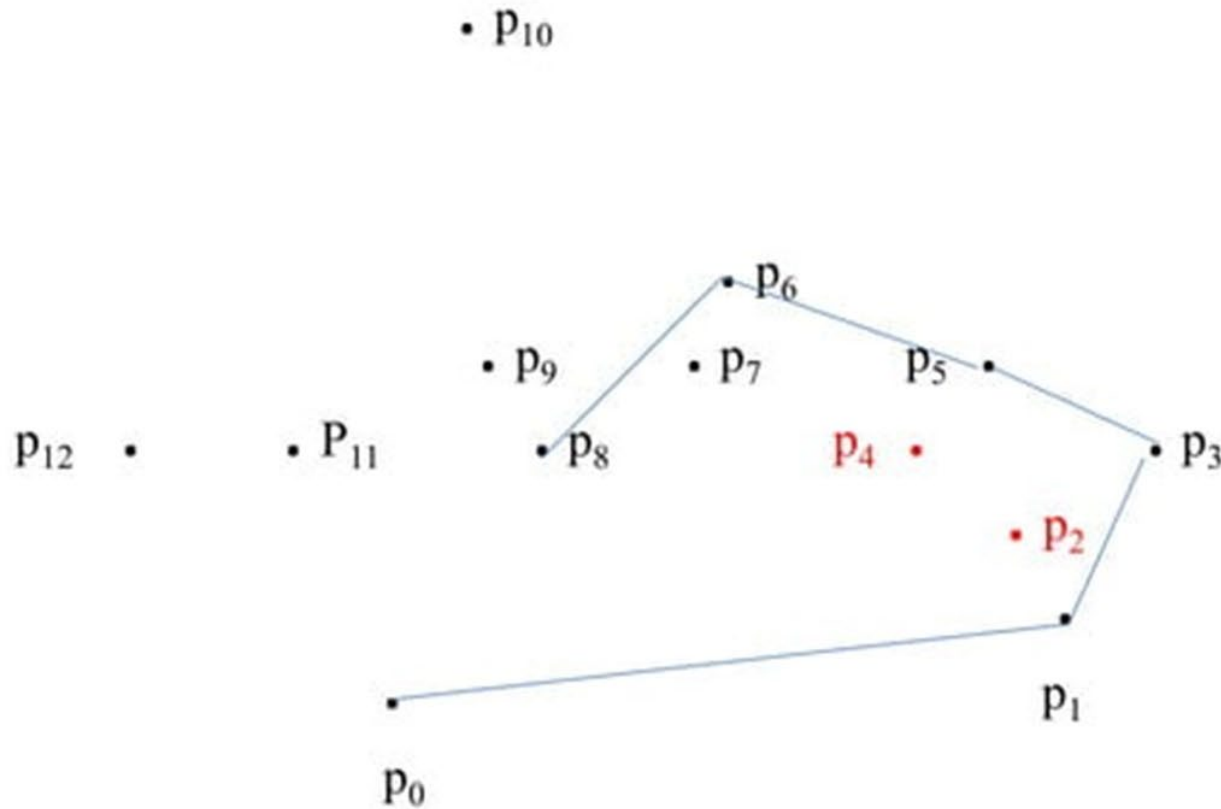
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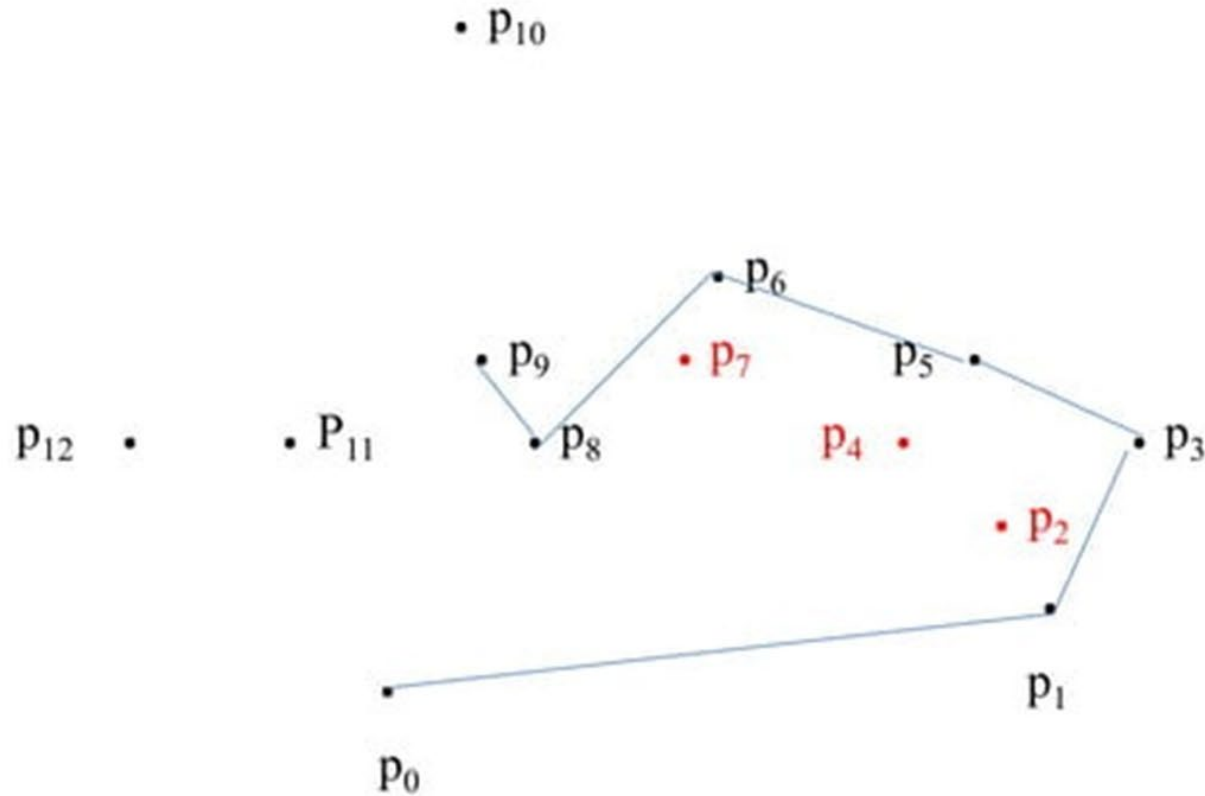
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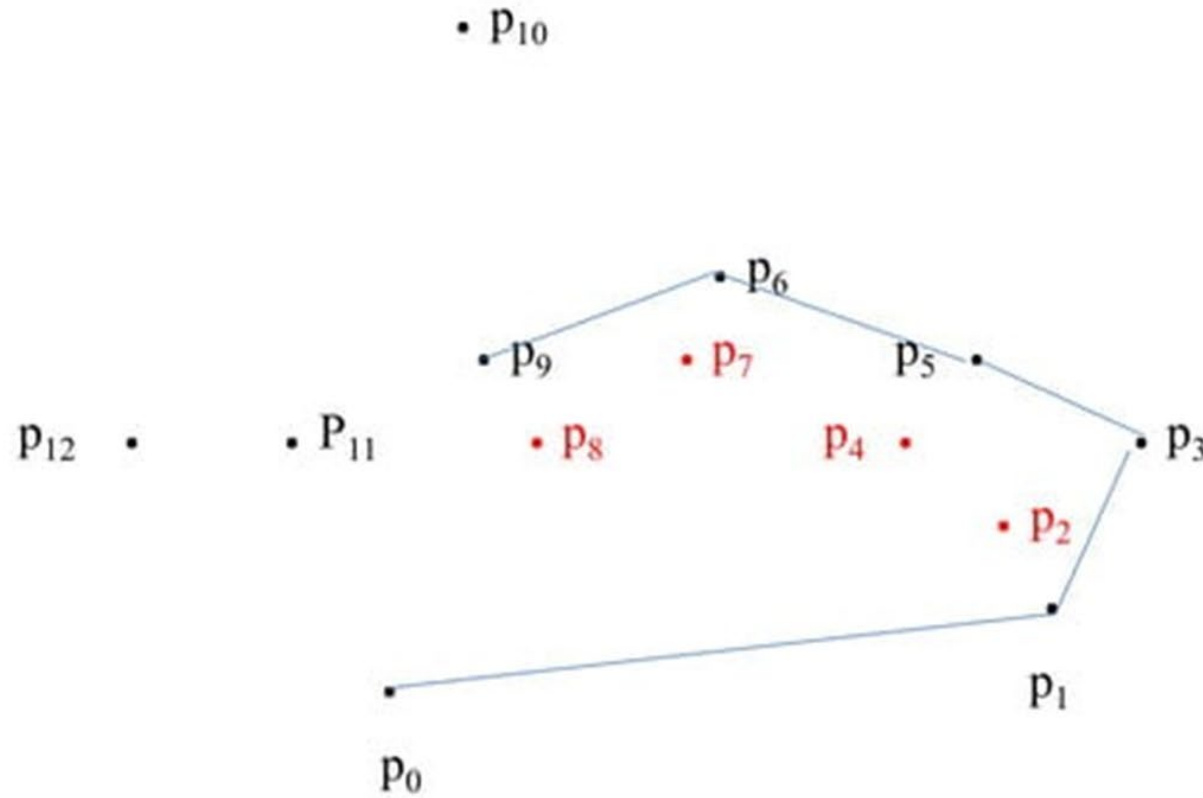
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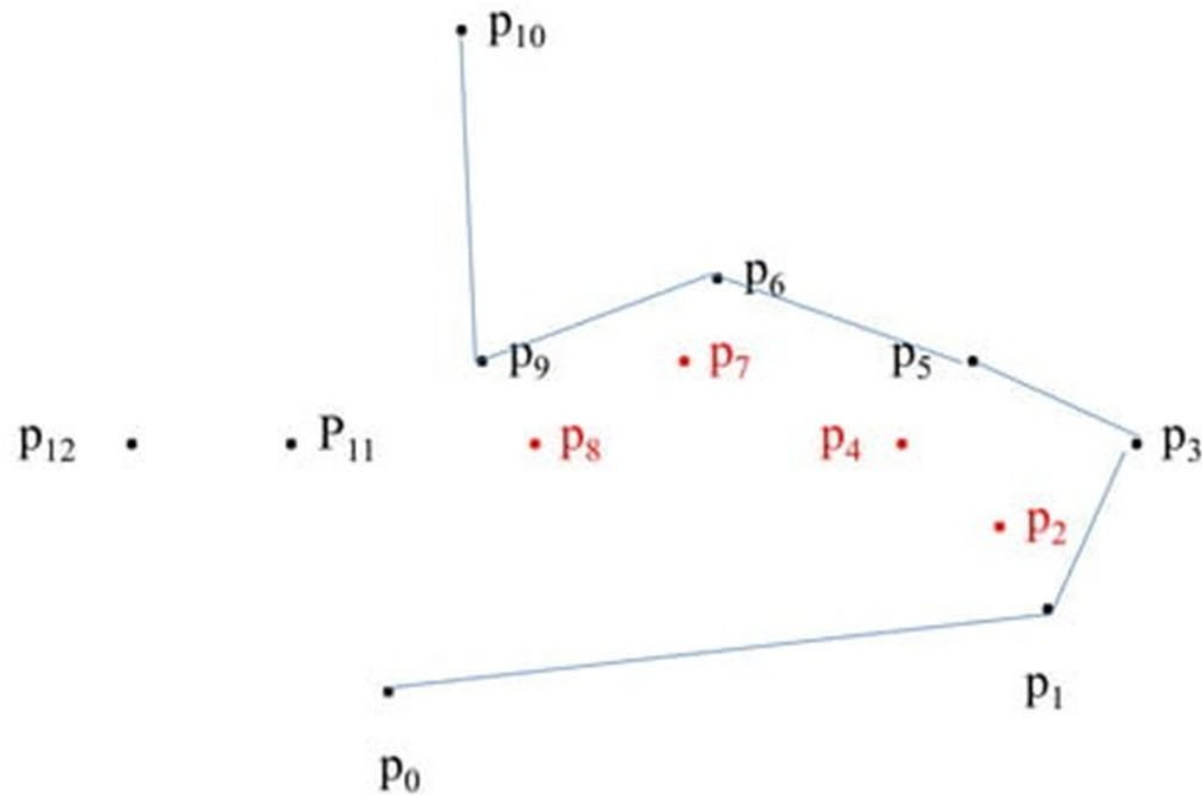
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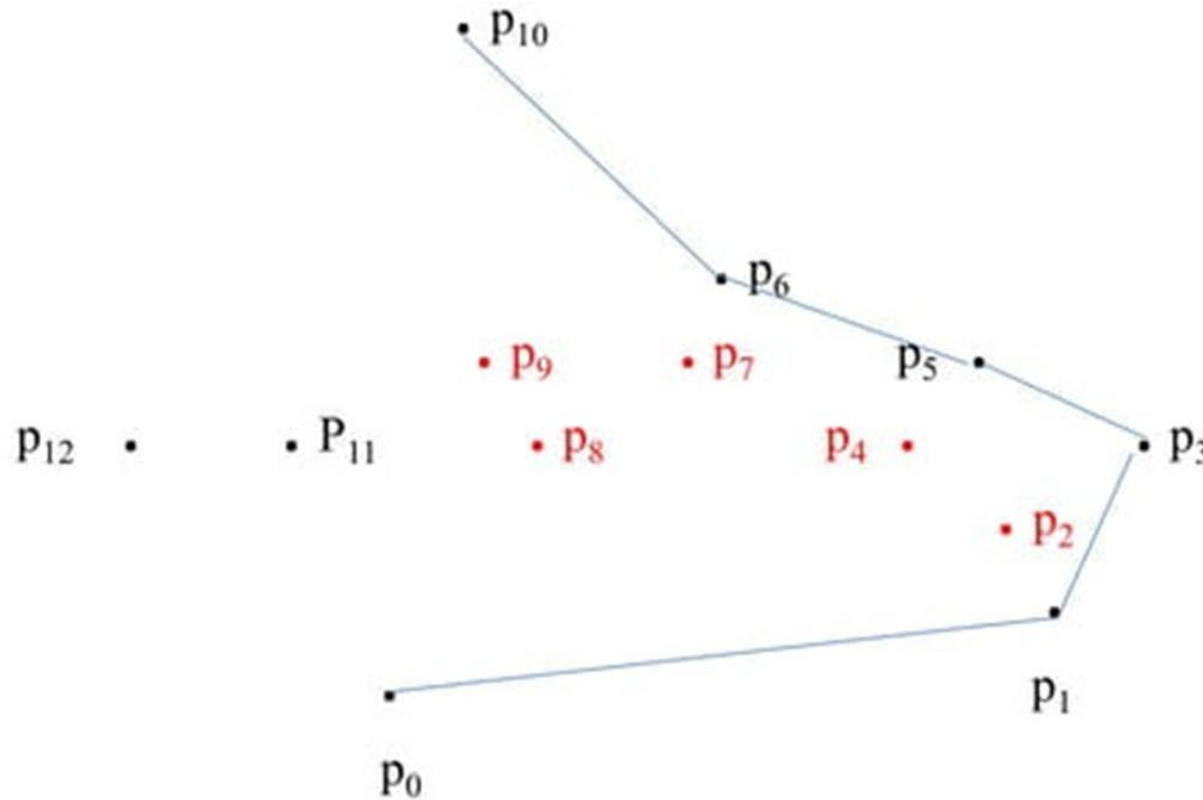
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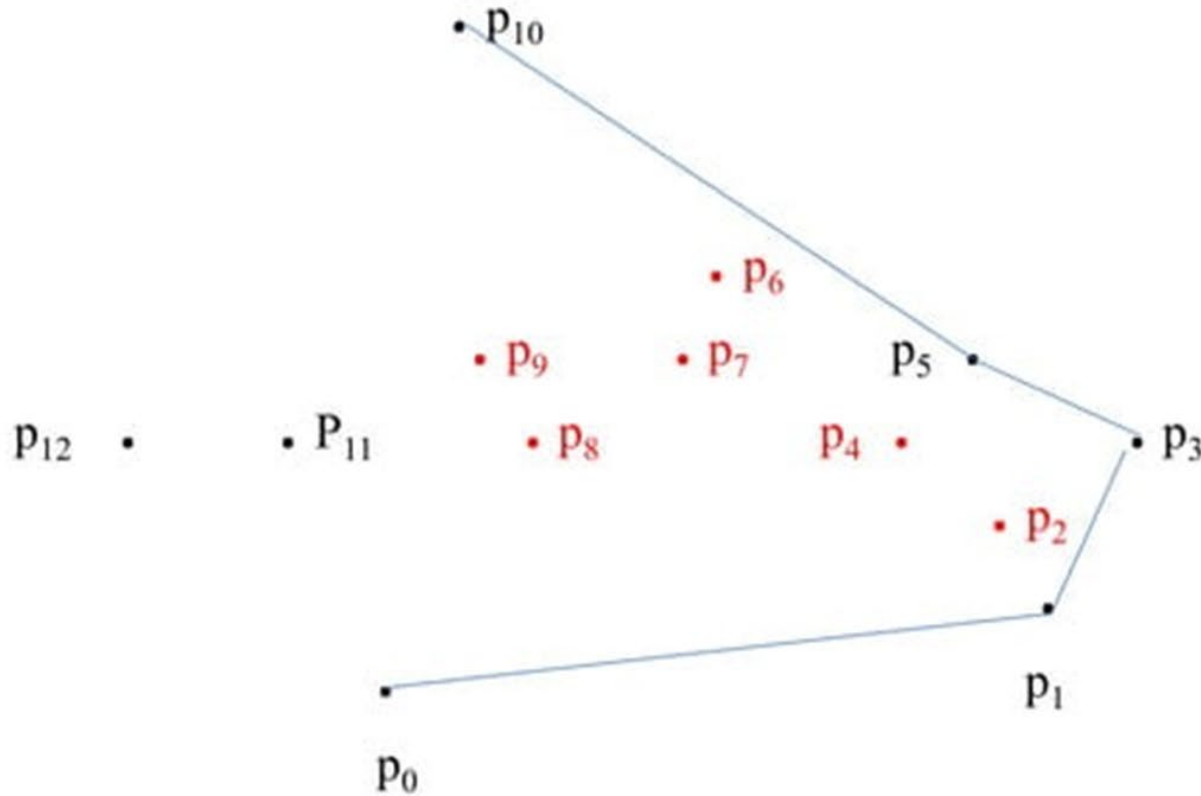
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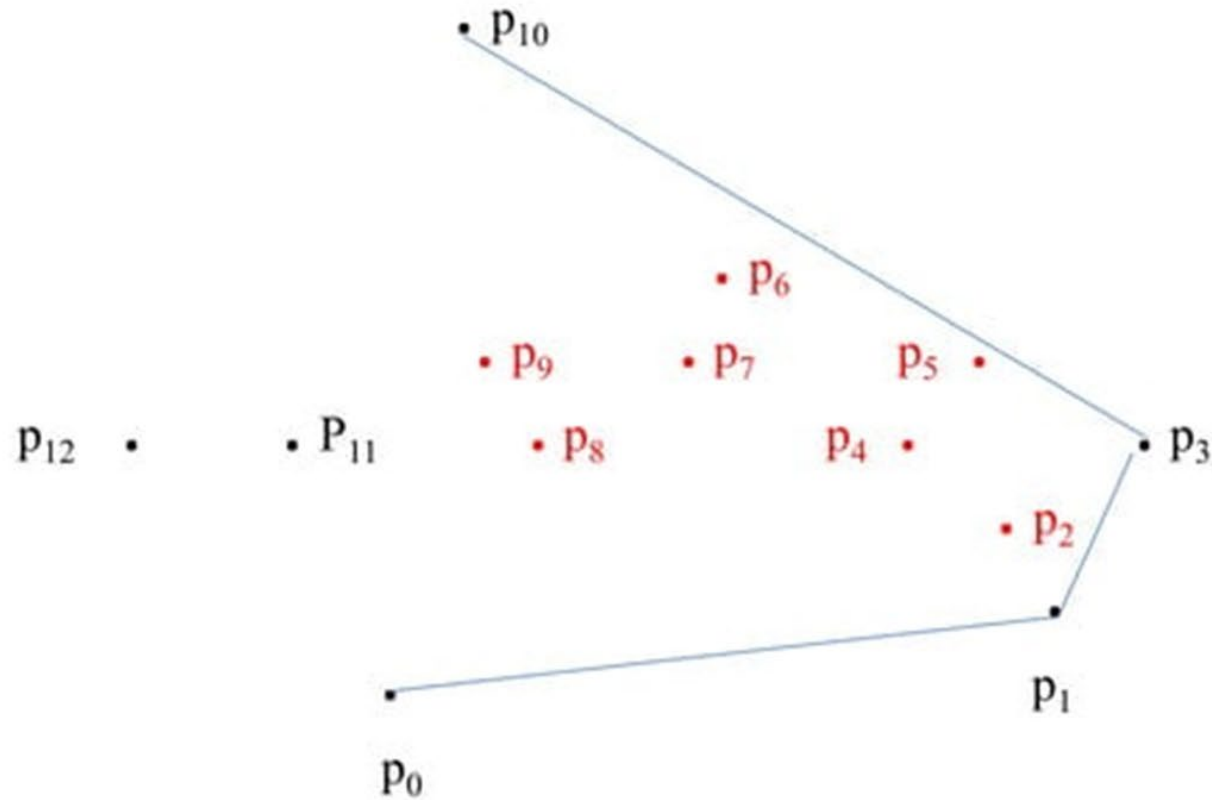
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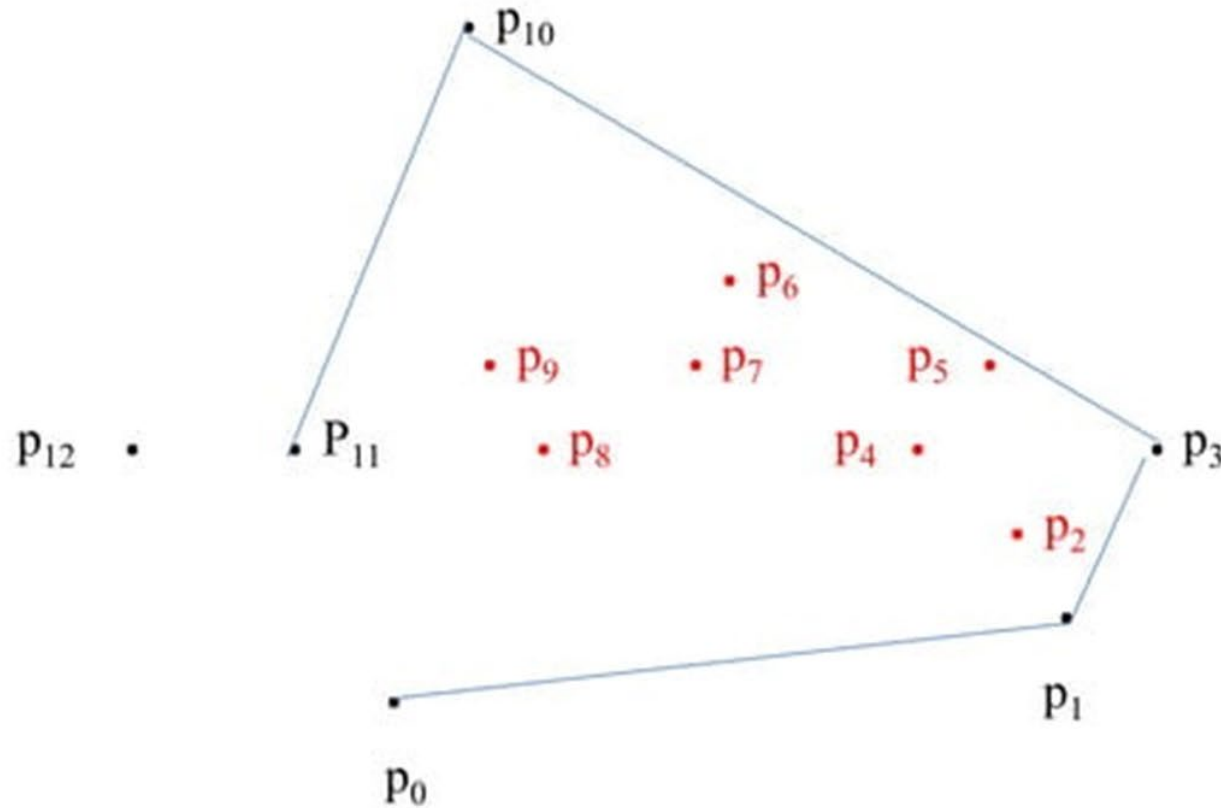
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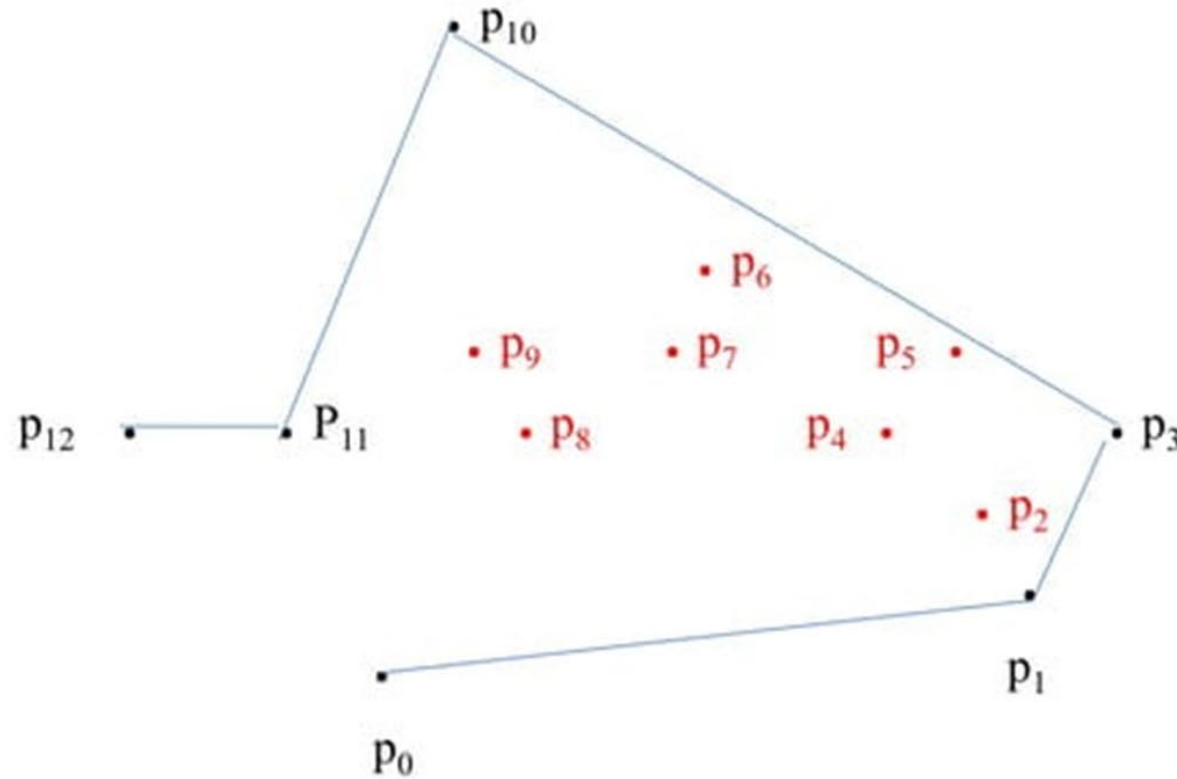
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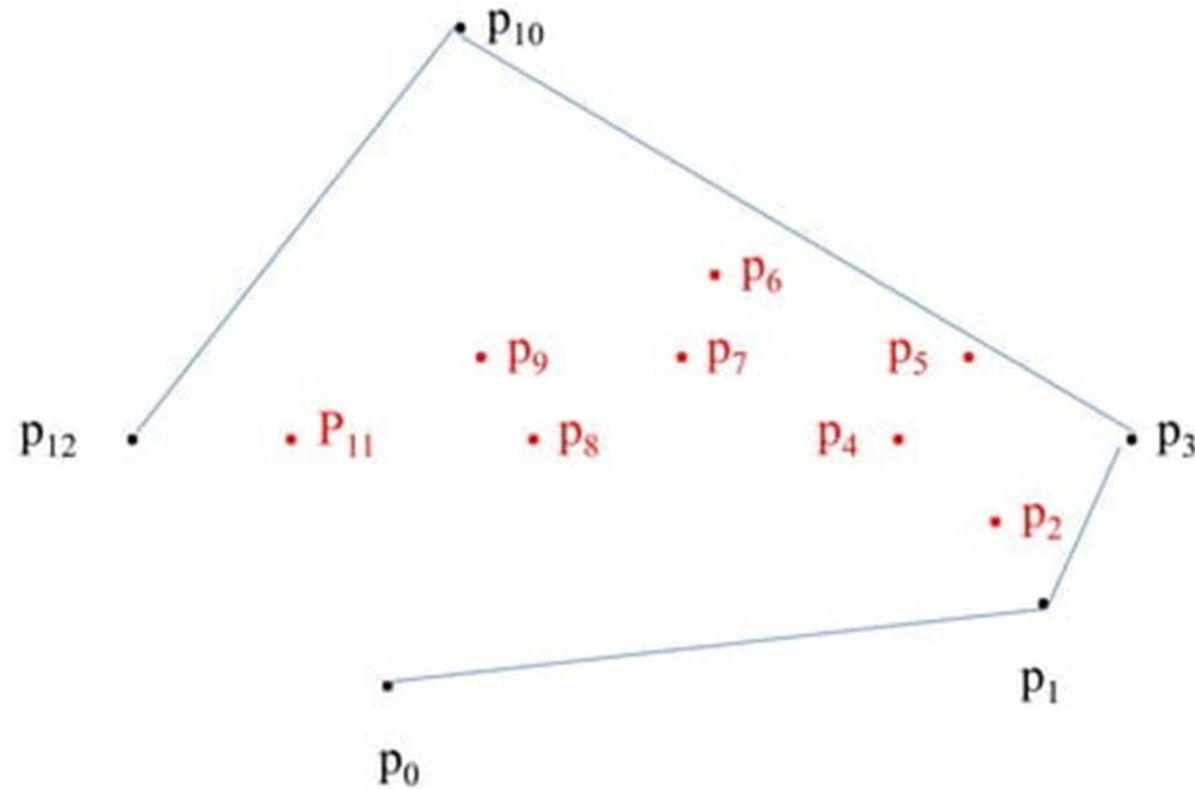
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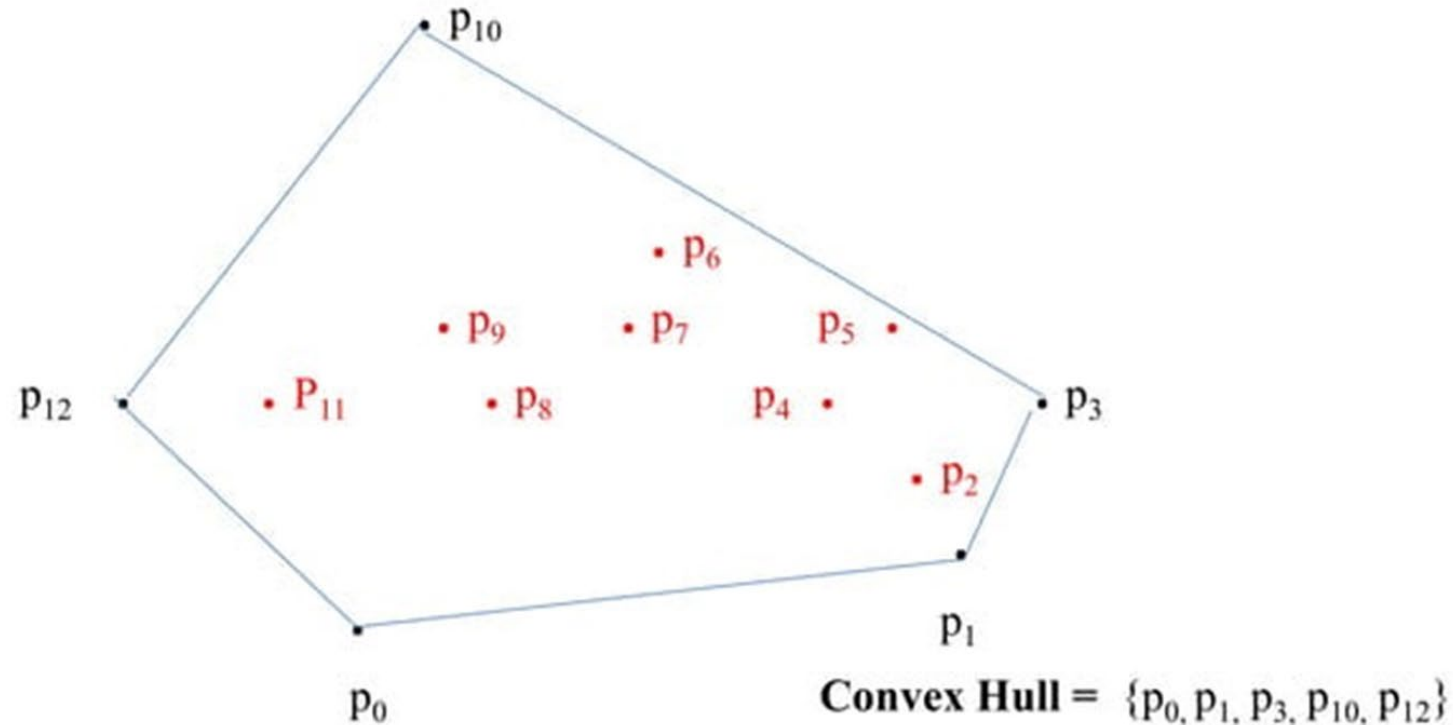
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Any

Question



PresenterMedia

Thank You!

**FOR YOUR
ATTENTION**

